

ASSIGNMENT 1: DATA ENGINEERING

Introduction to Apache Pig

Scenario: E-commerce Product Review Analysis

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1. Objective

This part processes an e-commerce product review dataset stored on HDFS using Apache Pig. The Pig Latin script loads reviews.csv, removes invalid ratings, groups the remaining reviews by product category, and computes the average rating of each category. The results are shown with DUMP and stored back on HDFS with STORE.

- Load review data from HDFS.
- Remove records where the rating is null.
- Remove records where the rating is below 1.0.
- Group valid reviews by product category.
- Calculate the average rating for each category.
- Display and store the category-wise averages.

2. Dataset

The file reviews.csv has no header row. A header would be treated as data by PigStorage. The fields are review_id, product_id, product_category, rating, and review_text.

```
1,P101,Electronics,4.5,Good product
2,P102,Electronics,3.8,Average product
3,P103,Books,4.2,Very useful
4,P104,Books,5.0,Excellent
5,P105,Clothing,3.5,Good quality
6,P106,Clothing,0.5,Poor
7,P107,Electronics,,No rating
8,P108,Books,4.8,Very good
9,P109,Clothing,2.5,Average
10,P110,Electronics,4.0,Good
```

Two rows are removed before aggregation:

review_id	Category	Rating	Reason
6	Clothing	0.5	Rating below 1.0
7	Electronics	empty / null	rating IS NULL

3. Pig Latin script (analysis.pig)

```
-- Load review dataset from HDFS
reviews = LOAD '/user/hadoop/reviews/reviews.csv'
USING PigStorage(',')
AS (
    review_id:int,
    product_id:chararray,
    product_category:chararray,
    rating:double,
    review_text:chararray
```

```
);

-- Remove null ratings and ratings below 1.0
non_null = FILTER reviews BY rating IS NOT NULL AND rating >= 1.0;

-- Group valid reviews by product category
category_group = GROUP non_null BY product_category;

-- Calculate average rating for each category
category_avg = FOREACH category_group GENERATE
    group AS product_category,
    ROUND_TO(AVG(non_null.rating), 2) AS average_rating;

-- Display the results
DUMP category_avg;

-- Store the results in HDFS
STORE category_avg
INTO '/user/aiml/category_average_rating'
USING PigStorage(',');
```

LOAD reads the CSV from HDFS. FILTER drops null ratings and ratings below 1.0. GROUP collects remaining reviews by product_category. AVG computes the mean rating in each group. ROUND_TO(..., 2) prints two decimal places. DUMP prints the tuples. STORE writes CSV output to /user/aiml/category_average_rating.

4. Commands used

```
export HADOOP_HOME=/usr/local/hadoop
export PATH=$HADOOP_HOME/bin:$HADOOP_HOME/sbin:$PATH
hdfs dfs -mkdir -p /user/hadoop/reviews
hdfs dfs -put reviews.csv /user/hadoop/reviews/
hdfs dfs -ls /user/hadoop/reviews
pig -x mapreduce analysis.pig
hdfs dfs -cat /user/aiml/category_average_rating/part-r-000000
```

5. Terminal screenshots

The figures below are live screenshots from the Hadoop cluster (Hadoop 3.3.6, Pig 0.17.0).

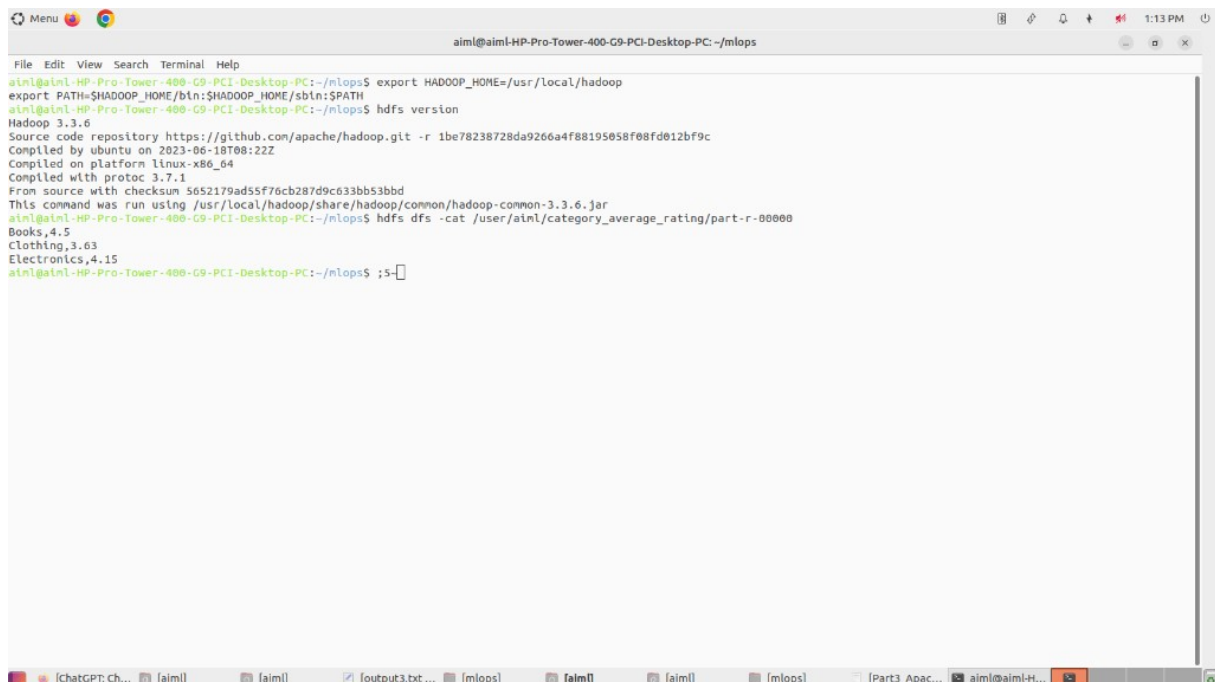


Figure 1. Hadoop version on the cluster.

```
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[.getmerge [-n] [-skip-empty-file] <src> <localdst>]
[.head <file>]
[.help [cmd ...]]
[.ls [-C] [-d] [-h] [-q] [-R] [-t] [-S] [-r] [-u] [-e] [<path> ...]]
[.mkdir [-p] <path> ...]
[.moveFromLocal [-f] [-p] [-l] [-d] <localsrc> ... <dst>]
[.moveToLocal <src> <localdst>]
[.mv <src> ... <dst>]
[.put [-f] [-p] [-l] [-d] [-t <thread count>] [-q <thread pool queue size>] <localsrc> ... <dst>]
[.renameSnapshot <snapshotDir> <oldName> <newName>]
[.rm [-f] [-r] [-R] [-skipTrash] [-safely] <src> ...]
[.rmdir [-ignore-fail-on-non-empty] <dir> ...]
[.setfacl [-R] [{-b|-k} {-m|-x <acl_spec>} <path>][[-set <acl_spec> <path>]]
[.setfacl [-n name [-v value] | -x name] <path>]
[.setrep [-R] [-w] <rep> <path> ...]
[.stat [format] <path> ...]
[.tail [-f] [-s <sleep interval>] <file>]
[.test [-defswrz] <path>]
[.text [-ignoreCrc] <src> ...]
[.touch [-a] [-n] [-t TIMESTAMP (yyyyMMdd:HHmmss)] [-c] <path> ...]
[.touchz <path> ...]
[.truncate [-w] <length> <path> ...]
[.usage [cmd ...]]

Generic options supported are:
- conf <configuration file> specify an application configuration file
- D <property=value> define a value for a given property
- fs <file:///hdfs://namenode:port> specify default filesystem URL to use, overrides 'fs.defaultFS' property from configurations.
- jt <local/ResourceManager:port> specify a ResourceManager
- files <file1,...> specify a comma-separated list of files to be copied to the map reduce cluster
- libjars <jar1,...> specify a comma-separated list of jar files to be included in the classpath
- archives <archive1,...> specify a comma-separated list of archives to be unarchived on the compute machines

The general command line syntax is:
command [genericOptions] [commandOptions]

Usage: hadoop fs [generic options] -ls [-C] [-d] [-h] [-q] [-R] [-t] [-S] [-r] [-u] [-e] [<path> ...]
aiml@aiml-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/mlops$ hdfs dfs -mkdir -p /user/hadoop/reviews/
hdfs dfs -put reviews.csv /user/hadoop/reviews/
hdfs dfs -ls /user/hadoop/reviews
Found 1 items
-rw-r--r-- 1 aiml supergroup 303 2026-09-25 13:23 /user/hadoop/reviews/reviews.csv
aiml@aiml-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/mlops$
```

Figure 2. Upload of reviews.csv to HDFS.

```
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[.stat [format] <path> ...]
[.tail [-f] [-s <sleep interval>] <file>]
[.test [-defswrz] <path>]
[.text [-ignoreCrc] <src> ...]
[.touch [-a] [-n] [-t TIMESTAMP (yyyyMMdd:HHmmss)] [-c] <path> ...]
[.touchz <path> ...]
[.truncate [-w] <length> <path> ...]
[.usage [cmd ...]]

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aiml@aiml-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/mlops$ hdfs dfs -mkdir -p /user/hadoop/reviews
hdfs dfs -put reviews.csv /user/hadoop/reviews/
hdfs dfs -ls /user/hadoop/reviews
Found 1 items
-rw-r--r-- 1 aiml supergroup 303 2026-09-25 13:23 /user/hadoop/reviews/reviews.csv
aiml@aiml-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/mlops$ pig -x mapreduce analysis.pig
2026-09-25 13:26:18,799 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-25 13:26:18,800 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-25 13:26:18,800 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-25 13:26:18,824 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-09-25 13:26:18,824 [main] INFO org.apache.pig.Main - Logging error messages to: /home/aiml/mlops/pig_1790322978821.log
2026-09-25 13:26:18,964 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file /home/aiml/.pigbootstrap not found
2026-09-25 13:26:18,999 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-25 13:26:18,999 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-25 13:26:19,312 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-analysis.pig-65a9feb5-5e74-40ce-b032-515f482255cc
2026-09-25 13:26:19,312 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-25 13:26:19,703 [main] WARN org.apache.pig.newplan.BaseOperatorPlan - Encountered Warning IMPLICIT_CAST_TO_DOUBLE 2 time(s).
2026-09-25 13:26:19,709 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,FILTER
2026-09-25 13:26:19,723 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-25 13:26:19,747 [main] INFO org.apache.pig.newplan.LogicalOptimizer.LogicalOptimizer - [RULES_ENABLED]=[addForEach, ColumnMapKeyPrune, ConstantCalculator, Group
ByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PushDownF
```

Figure 3. Running pig -x mapreduce analysis.pig.

```

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Job Stats (time in seconds):
JobId  Maps  Reduces  MaxMapTime  MinMapTime  AvgMapTime  MedianMapTime  MaxReduceTime  MinReduceTime  AvgReduceTime  MedianReduceTime  Alias  Featu
re  Outputs
job_local1473589168_0001  1  1  n/a  n/a  n/a  n/a  n/a  n/a  category_avg,category_group,non_null,reviews  GROUP_BY,COMB
INNER  hdfs://localhost:9000/tmp/temp1504391494/tmp2137620816,

Input(s):
Successfully read 18 records (11512170 bytes) from: "/user/aiml/reviews.csv"

Output(s):
Successfully stored 3 records (11511636 bytes) in: "hdfs://localhost:9000/tmp/temp1504391494/tmp2137620816"

Counters:
Total records written : 3
Total bytes written : 11511636
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local1473589168_0001

2026-09-25 13:26:26,126 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-25 13:26:26,127 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-25 13:26:26,127 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-25 13:26:26,131 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - Success!
2026-09-25 13:26:26,133 [main] WARN org.apache.pig.data.SchemaTupleBackend - SchemaTupleBackend has already been initialized
2026-09-25 13:26:26,137 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-09-25 13:26:26,137 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
(Books,4.67)
(Clothing,3.0)
(Electronics,4.1)
2026-09-25 13:26:26,166 [main] WARN org.apache.pig.newplan.BaseOperatorPlan - Encountered Warning IMPLICIT_CAST_TO_DOUBLE 2 time(s).
2026-09-25 13:26:26,167 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textou
tputformat.separator
2026-09-25 13:26:26,170 [main] ERROR org.apache.pig.tools.grunt.Grunt - ERROR 6000: <file analysis.pig, line 33, column 0> Output Location Validation Failed for: '/user/aiml
/category_average_rating More info to follow:
Output directory hdfs://localhost:9000/user/aiml/category_average_rating already exists
Details at logfile: /home/aiml/mllops/pig_1790322978021.log
2026-09-25 13:26:26,184 [main] INFO org.apache.pig.Main - Pig script completed in 7 seconds and 412 milliseconds (7412 ms)
aiml@aiml-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/mllops$

```

Figure 4. Pig DUMP output and job summary.

```

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aiml@aiml-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/mllops$ export HADOOP_HOME=/usr/local/hadoop
aiml@aiml-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/mllops$ export PATH=$HADOOP_HOME/bin:$HADOOP_HOME/sbin:$PATH
aiml@aiml-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/mllops$ hdfs version
Hadoop 3.3.6
Source code repository https://github.com/apache/hadoop.git -r 1be78238728da9266a4f88195858f08fd012bf9c
Compiled by ubuntu on 2023-06-18T08:22Z
Compiled on platform linux-x86_64
Compiled with protoc 3.7.1
From source with checksum 5652179ad55f76cb287d9c633bb53bbd
This command was run using /usr/local/hadoop/share/hadoop/common/hadoop-common-3.3.6.jar
aiml@aiml-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/mllops$ hdfs dfs -cat /user/aiml/category_average_rating/part-r-00000
Books,4.5
Clothing,3.63
Electronics,4.15
aiml@aiml-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/mllops$ hdfs dfs -cat /user/aiml/category_average_rating/part-r-00000 >output.txt
aiml@aiml-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/mllops$

```

Figure 5. Reading stored output from HDFS.

6. Results

Valid ratings: Books = {4.2, 5.0, 4.8}; Clothing = {3.5, 2.5}; Electronics = {4.5, 3.8, 4.0}. DUMP output from Figure 4: (Books,4.67), (Clothing,3.0), (Electronics,4.1).

Category	Valid reviews	Sum	Average
Books	3	14.0	4.67
Clothing	2	6.0	3.0
Electronics	3	12.3	4.1

STORE reported that /user/aiml/category_average_rating already existed, so that directory was not overwritten. The DUMP values above are the results of this run. To store again, remove the old path with:
hdfs dfs -rm -r /user/aiml/category_average_rating

7. Discussion

Why is GROUP used? It puts reviews with the same category into one bag so AVG can run once per category.

Why filter nulls and ratings below 1.0? Nulls are missing scores. Ratings below 1.0 are invalid. Both would distort the mean.

What does AVG() do? It computes the arithmetic mean of rating values inside each category group.

What is PigStorage(',')? It tells Pig that input and output fields are separated by commas.

Where is the data stored? Input: /user/hadoop/reviews/reviews.csv on HDFS. Output: /user/aiml/category_average_rating on HDFS.

8. Conclusion

Apache Pig loaded the review CSV from HDFS, removed invalid ratings, grouped reviews by category, and computed averages of 4.67 (Books), 3.0 (Clothing), and 4.1 (Electronics). The terminal screenshots show the HDFS upload, the MapReduce run, and the DUMP output.